

Problem

A three-phase line has an impedance of $1 + j3 \Omega$ per phase. The line feeds a balanced delta-connected load, which absorbs a total complex power of $12 + j5$ kVA. If the line voltage at the load end has a magnitude of 240 V, calculate the magnitude of the line voltage at the source end and the source power factor.

Step-by-step solution

Step 1 of 3

The line feeds delta connected load. In delta connection, line voltage and phase voltage are same.

$$\begin{aligned} V_p &= V_L \\ &= 240 \text{ V} \end{aligned}$$

Write the expression for total complex power.

$$S_T = 3V_{ph} I_{ph}^*$$

The total complex power is,

$$\begin{aligned} S_T &= 12 + j5 \text{ kVA} \\ &= 13 \angle 22.62^\circ \text{ kVA} \end{aligned}$$

Calculate the phase current.

$$\begin{aligned} I_{ph}^* &= \frac{S_T}{3V_{ph}} \\ &= \frac{13 \times 10^3 \angle 22.62^\circ}{(3)(240)} \\ &= 18.055 \angle 22.62^\circ \text{ A} \\ I_{ph} &= 18.055 \angle -22.62^\circ \text{ A} \end{aligned}$$

Step 2 of 3

Calculate the line current.

$$\begin{aligned} I_L &= \sqrt{3} I_{ph} \angle -30^\circ \\ &= \frac{18.055 \sqrt{3} \angle (-22.62 - 30)^\circ \text{ A}}{\angle -30^\circ} \\ &= 31.27 \angle -52.62^\circ \text{ A} \end{aligned}$$

Calculate the source line voltage.

$$\begin{aligned} V_{L(\text{source})} &= I_L Z_{\text{line}} + V_L \\ &= (31.27 \angle -52.62^\circ)(1 + j3) + 240 \\ &= 93.53 + j32.1 + 240 \\ &= 333.53 + j32.1 \\ &= 335.07 \angle 5.49^\circ \text{ V} \end{aligned}$$

Therefore, the value of source line voltage is $\boxed{335.07 \angle 5.49^\circ \text{ V}}$.

Step 3 of 3

Calculate the source power factor.

$$\begin{aligned} pf &= \cos(\theta_{V_{L(\text{source})}} - \theta_{I_L}) \\ &= \cos(5.49^\circ - (-52.62^\circ)) \\ &= \cos(58.11^\circ) \\ &= 0.528 \text{ lagging} \end{aligned}$$

Therefore, the power factor of the source is 0.528 lagging.